

The repetition ratio r/R measures length and encoding, not linguistic status: a reanalysis of Sproat (2014)

Bharath Rao
Independent

June 2026

Abstract

The repetition ratio rule (RRR), based on the ratio r/R of geminate to total repeats, has been proposed as a discriminant between linguistic and non-linguistic symbol systems. We show that for short and moderate-length inscriptions the expected r/R (precisely, the ratio of expectations $E[r]/E[R]$) under a random control is $2/L$, where L is the per-inscription token count. The threshold $r/R = 0.10$ therefore functions as a test of mean inscription length ≥ 20 . We demonstrate that RRR misclassifies in both directions when length is matched: top-100 American and Finnish given names, exchange-traded commodity labels, and 10,749 deciphered Mesopotamian seal legends ($r/R \in [0.13, 0.53]$) are misclassified as non-linguistic; ISBN-13 lists, IPv4 addresses, cryptographic hashes, and character-level chess game records ($r/R \in [0.0004, 0.10]$) are misclassified as linguistic. We show that it fails as a discriminant because the premises implicit in its construction are unsound: it is biased against gemination, against particular encoding depths, and against short inscriptions. Varying length or encoding of random strings flips the result. We conclude that r/R does not discriminate language from non-language, and distill its failure modes into six requirements that any proposed discriminant must satisfy before its verdicts can be trusted.

1 Introduction

Whether statistical measures can distinguish writing from non-linguistic symbol systems without decipherment is a long-standing question, debated most prominently over the Indus Valley corpus: Rao et al. (2009); Rao (2010) argued from bigram conditional entropy and block entropy that undeciphered inscriptions can cluster with known writing systems, against the thesis that the symbols are not writing (Farmer, Sproat, and Witzel, 2004). The methodological debate continued in *Computational Linguistics* (Sproat, 2010a; Lee, Jonathan, and Ziman, 2010b; Rao et al., 2010; Sproat, 2010b). Sproat (2014) assembled a larger and more balanced comparison set, found the entropic measures unable to separate the classes, and proposed a new general-purpose discriminant: the ratio r/R of *adjacent* symbol repetitions to *total* symbol repetitions, summed across inscriptions. A threshold of $r/R = 0.10$ separates the linguistic systems below this value from the non-linguistic systems above it (henceforth repetition ratio rule = RRR). We evaluate this proposal as a classifier in its own right, independently of any particular undeciphered corpus.

We present a structural and empirical falsification of the r/R verdict by showing:

1. r/R is effectively a proxy for inscription length and simply tests whether the mean inscription length is ≥ 20 . More precisely, $E[r/R] \rightarrow 2/L$ for texts of moderate length ($L \ll V$) and $E[r/R] \rightarrow 1/V$ for texts of very long length ($L \gg V$). The threshold $r/R = 0.10$ corresponds to $L = 20$ in the former and $V = 10$ in the latter, where L is the average inscription length and V is the symbol inventory size.

2. A set of modern linguistic and non-linguistic corpora produce r/R values that cross the 0.10 threshold in both directions despite having known linguistic status. The overlap between the “linguistic” and “non-linguistic” clusters is large enough that the $r/R = 0.10$ threshold loses its discriminative power on any corpus set that includes them.
3. The metric is sensitive to tokenization-unit choice (encoding-depth) and allography. For unknown symbol systems such as the Indus script, these are exactly what decipherment would establish, so the r/R verdict is not a viable discriminant.

2 The r/R measure and the Repetition Ratio Rule (RRR)

For a sequence $s_1, s_2, \dots, s_L$ of symbol tokens, define $X_i = \mathbf{1}[s_i = s_{i-1}]$ and $Y_i = \mathbf{1}[s_i \in \{s_1, \dots, s_{i-1}\}]$ for $i \geq 2$. Then $r = \sum_i X_i$ counts *geminate* (adjacent) repetitions and $R = \sum_i Y_i$ counts *all* repetitions. The values of r and R are computed per inscription, then are summed over the corpus and ratio of the two sums is computed.

The three elements of the formula encode three premises:

- Repetition is a characteristic of linguistic systems: writing systems reuse a finite stock of letters/syllables/morphemes, so $R > 0$ for any text of non-trivial length. In other words, strings of non-repeating symbols are more likely to be non-linguistic.
- Geminate repetition is a characteristic of decorative or ritual systems: heraldic emblems, totem-pole figures, and Mesopotamian deity symbols on kudurrus are often deliberately doubled adjacently. Language avoids adjacent repetition because function words like English *the* and *is* rarely occur back-to-back, so r remains small while R grows.
- An inscription with zero repetitions does not contribute to a linguistic signal. Per-inscription scoping (sum-then-ratio across inscriptions) accumulates the linguistic signal from each individual inscription. Cross inscription repetitions do not contribute to the linguistic signal.

The r/R repetition ratio as a linguistic-system discriminant is not derived from any general model of language or orthography. The three premises appear to be post-hoc generalizations from the observed properties of the Indus corpus: short inscriptions with few repetitions, most of which are geminate. In effect, the Indus corpus is treated as an exemplar of non-linguistic symbol systems, and RRR scores any corpus by its resemblance to that exemplar.

The derivation of the discriminant threshold of $r/R = 0.10$ is unclear and seems to be empirically determined based on a set of samples of different linguistic and non-linguistic texts. We compute for it a Youden’s J of 0.857 (§4.3) on the corpora with known linguistic status, so it is a strong discriminant for the analyzed data sets for the purpose of binary sorting of these sets. Throughout, we apply the rule “linguistic iff $r/R < 0.10$,” so corpora at exactly 0.10 (Egyptian, Asian emoticons) class as non-linguistic. All but two of the RRR linguistic comparanda fall below the threshold and most non-linguistic comparanda fall above (in Sproat’s own terms, Sumerian and the kudurrus are misclassified; Egyptian and Asian emoticons borderline).

2.1 Analysis of the premises

Repetition is chosen as the linguistic signal to be measured since it is seen as a fundamental quality of language. An absence of repetition across all inscriptions should first prompt more sampling. The per-inscription expectation $E[r/R]$ proper is not even defined when zero repetitions is probable. The r/R ratio therefore, is indeterminate when $R = 0$. If the sample size is already large (say 1,000 glyphs or 100 inscriptions) with no repetitions at all, then the corpus is better classed as non-linguistic. This is reasonable because allowing $R = 0$ to mean language would imply that any non-repeating sequence, such as the images from a kaleidoscope, would

be classed as language. We can adjust the formula to avoid the division by zero by changing it to $\frac{r+\epsilon}{R+\epsilon}$ for an arbitrarily small $\epsilon > 0$: this leaves all defined values essentially unchanged and assigns $r/R = 1$ to corpora with zero repetitions, classing them as non-linguistic. It also makes the $r = R = 0$ edge case explicit rather than indeterminate.

2.1.1 The gemination penalty r

The gemination penalty seems to be derived from the observation of the power law of word tokens in English, where articles and prepositions peak. This penalty seems to be premising that the encoding of the symbol system is at the level of words. Word-level reduplication is a normal and broadly observed phenomenon in many languages. About 85% of the 368 languages in Rubino’s WALS survey show productive reduplication (Inkelas, 2014), marking for example plurality (Indonesian *anak-anak*), intensification (Italian *grande grande*), and frequentatives (Hawaiian *holoholo*). Many of these languages also reduplicate at the syllable level.

Syllable reduplication is common in languages such as Sanskrit and Greek due to morphological rules such as the reduplication of the initial syllable in the perfect tense.¹ Sanskrit in particular has copious adjacent syllable repetition, including triples. The word *dadad* for example is attested six times in the Rigveda, twice in verse 10.85.41 (*somo dadad gandharvāya gandharvo dadad agnaye*).² A regex search over a digitization of the Monier-Williams dictionary (Monier-Williams, 1899) finds $\approx 2,700$ headwords containing an adjacent repeated consonant–vowel sequence (*dada-*, *mama-*, ...), ≈ 580 of them word-initial.³ Syllable repetition is rare in English but can be seen in words like “tartar, murmur, testes” etc.

At the letter level, gemination is far more prevalent in English, with about 21.9% of words containing a doubled letter.⁴ Doubling glyphs to represent geminated consonants or long vowels is an orthographic convention. German compounds can produce even longer runs, for example *Schiffahrt*. Japanese on the other hand uses a small *tsu* (sokuon, っ or っ) to indicate gemination. Some languages use doubled letters to denote long vowels while others may use a diacritic such as a macron. The gemination penalty therefore arbitrarily punishes a certain orthography rather than any non-linguistic feature. So Devanāgarī encoded मम gets a gemination penalty and is classified as maximally non-linguistic at 1.0, while the same word when IAST encoded as *mama* gets a maximally linguistic score of 0.0.

Languages like Finnish which use geminates to indicate long vowels and have geminated consonants pay an extra large penalty. The top-100 Finnish given names used in this paper contain 89 geminate letter pairs per 1,000 letters, against 29 per 1,000 for the top-100 American names and 17 per 1,000 glyph tokens for the full ICIT Indus corpus (4,640 inscriptions, 16,741 glyph tokens); Finnish orthography is therefore exposed to the r penalty roughly five times more heavily than the Indus corpus itself.⁵ On the flip side, purely random sequences are non-linguistic but contain adjacent repeats only at the chance rate $1/V$, and sequences sampled without replacement (shuffled cards) contain none at all. Representations of shuffled cards and cryptographic hashes are therefore barely penalized by the r value. This noise is ameliorated

¹Prescribed in Pāṇini’s *Aṣṭādhyāyī* (Ashtadhyayi.com, 2024): *liṭi dhātor anabhyāsasya* (A. 6.1.8) ordains the doubling in the perfect, and subsequent rules such as A. 6.1.17 govern the shape of the reduplicate.

²Token counts over the *sāṇihitā* text; searchable at <https://www.shrauta.com>. The Atharvaveda adds five further *dadad* tokens inside sandhi compounds, two of them quoting this same mantra.

³Pattern `([ˆaeiou])(.)\1\2` applied to the SLP1-encoded headwords of the MW digitization served by <https://www.shrauta.com> (`c=mw`).

⁴9,461 of the 43,220 words in the bundled SCOWL American-English list; computed by `compute_rR.py`.

⁵Name-list rates computed by `compute_rR.py` from the lists used throughout this paper. The Indus rate is computed from the sign-sequence export of the Interactive Corpus of Indus Texts (ICIT, Wells & Fuls, <https://www.indus.epigraphica.de>) after expanding the 31 ICIT sign codes that encode doubled or quadrupled glyph groups into their constituent glyphs (mapping bundled as `icit_xlits.csv`) and treating damage and space markers as sequence breaks. Further allographic normalization (merging variant forms of the same sign) raises the Indus rate to ≈ 28 per 1,000; the Finnish multiple is three- to five-fold under any treatment.

in longer inscriptions since gemination as a proportion of the total text is often negligible.

2.1.2 The encoding penalty R

The total repetition measure R is essentially a bias to certain kinds of encoding depth in addition to orthography. Some orthographies are closed systems and have a finite number of glyphs. These include alphabetic, abjads and abugida systems. In such systems new symbols do not need to be invented, but are often done so as allographs for aesthetic or cultural reasons. Other systems may be logosyllabic, like Chinese Hanzi and Japanese Kanji, which can invent new glyphs indefinitely. The total number of Chinese Hanzi glyphs has no fixed upper bound; as of 2025 more than 100,000 distinct Hanzi have been identified and encoded in the CJK Unified Ideographs blocks (base block plus extensions) of the Unicode Standard (The Unicode Consortium, 2024). Historically, new Hanzi and Kanji glyphs would be created as needed and would gain traction via a network effect facilitated by inclusion in important books. In modern times, the modern administrative apparatus facilitates the recognition of new glyphs. The repetition of a glyph depends on the length of the inscription, the encoding depth and the symbol inventory. The larger the inventory, the smaller the chances of repetition. For example, Sumerian has around a dozen glyphs for the same sound and different ones are used for different words and often the same word can be written differently.⁶ In Sumerian *lugal* is attested in 6 or more ways for example  or .⁷ Cuneiform is polyvalent, so not only can different glyphs be used for the same sound, they can be read as different sounds depending on context. So the phonemic or syllabic repetition in the spoken language may not appear in the written form.

Chinese on the other hand, typically encodes each morpheme with a single glyph, and repeated glyphs in short inscriptions are unlikely. The 308 subway station names of Shenzhen for example, have zero stations with repeating glyphs, while Beijing’s 250 stations have only 2 station names with repeated glyphs. These stations class as linguistic for pinyin encoding, showing that it’s the encoding that matters and not the linguistic nature of what is written.

One of the reasons for the scoring to be highly unstable is the asymmetric encoding of spacing. Most of the corpora analyzed take space as a symbol. However, old inscriptions did not have spacing or punctuation. If a space as a symbol can be defined as a blank area separation between two other symbols with the approximate width of visible symbols, then barn stars should have a space between each star and would score 0.0, which is maximally linguistic. The same is true for weather icons, which are often organized in a table separated by gridlines. The R value therefore, is unstable based on not just the depth of encoding, but also the fidelity of encoding.

2.1.3 The brevity penalty due to per inscription reset

The r and R measures are counted per inscription. This design leads to expected effects such as lone English words scoring non-linguistic while multiple words score as linguistic. So short lone words don’t tack on any signal to the ratio. In fact, every word in the sentence “*So short lone words don’t tack on any signal to the ratio*” contributes zero to the signal if they are by themselves. However together as a sentence, they will score as linguistic. This outcome seems to be the central point of the ratio: linguistic inscriptions are long and therefore they have repeats. This assumption is not universally true. Short linguistic inscriptions abound: place names such as the subway stations mentioned above, personal names, titles, commodity labels, and status indicators such as “new” or “sold”. Indeed, bronze age seals themselves

⁶E.g. the reading /du/: ePSD2 attests du, du₃, du₅, du₆, du₇, du₈, du₉, du₁₀, du₁₁, du₁₂, du₁₄, and du₂₆ — twelve distinct signs sharing one reading, conventionally distinguished by subscript (ePSD2 Project, 2017).

⁷The ePSD2 entry for *lugal* lists six base spellings and 138 distinct attested written forms across 49,910 instances (ePSD2 Project, 2017).

typically are short inscriptions, often bearing only the name of the owner, occasionally with their office.⁸ And the discriminant verdict on real, deciphered bronze-age seals is not hypothetical. Extracting every transliterated seal legend from the CDLI corpus — 10,749 distinct legends, mean length 9.3 signs, readable name-title-patronymic formulae — and scoring them as a corpus gives $r/R = 0.134$, 95% CI [0.127, 0.142]: actual cuneiform writing, classified non-linguistic, with the entire confidence interval above the threshold. The gemination rate of these legends is only 11 per 1,000 signs — lower than the Indus corpus — so the misclassification is driven purely by brevity. Modern commodity labels make the point concretely: the 58 commodities traded on the major exchanges (wheat, coffee, cotton, copper, rubber, ...) score $r/R = 0.169$ — non-linguistic — with a gemination rate of 39 per 1,000 letters, higher than the American name list. The labels of trade goods, the very content that administrative seals and tags are believed to record, fail the test.⁹ Resetting the count at every inscription zeroes out the linguistic signal of the most prevalent writing forms on short length inscriptions. The cross-inscription recurrence that *is* the actual linguistic signal (the productive recurrence of 南, 园, 湾, 湖 across hundreds of Shenzhen station names) is invisible to the per-text scoping.

While the ratio does score long language strings as linguistic, it also scores purely random strings as linguistic with increased length. This is simply because random strings from a closed inventory of symbols rarely contain adjacent repeats but, at sufficient length, contain plenty of non-adjacent repeats.

3 Mathematical reduction: $r/R \approx 2/L$

To mathematically formalize the behavior of RRR for linguistic vs non-linguistic distributions, we need a distribution that has no structure of any kind as a control to evaluate failure modes.

3.1 The null distribution

Let us consider a corpus which has an inventory of V distinct symbols. Let L be the per-inscription token count, and let $s_1, \dots, s_L$ be i.i.d. uniform on $\{1, \dots, V\}$. Since each symbol is drawn from the alphabet uniformly and independently, this is essentially a random distribution with no linguistic or other structure except what can occur by pure chance. Such a distribution serves as a null hypothesis model to analyze the behavior of the ratio on a control distribution that has no inherent structure. For brevity, we refer to this i.i.d. uniformly random-sampled null hypothesis distribution as the *null distribution*; to the expected r/R it predicts at given corpus parameters (L, V) , computed in closed form below, as the *null ratio value*; and to that plot of r/R values traced as a function of length under the null distribution as the *null curve*. RRR scores of real corpora are compared against this control distribution. A lower score suggests structured suppression of repetition due to various linguistic phenomena such as phonotactics and morphology or encoding and orthographic rules. A higher score suggests structured enhancement such as geminate phonology, low- V encoding or simply brevity. We compute $E[r/R]$ under this null distribution in closed form below.

Throughout this paper we take the slight notational liberty of writing $E[r/R]$ for the ratio of expectations $E[r]/E[R]$. RRR computes $\sum_m r_m / \sum_m R_m$ over the M inscriptions of a corpus, and by the law of large numbers this statistic converges to $E[r]/E[R]$ as M grows.

⁸E.g. the Babylonian seals published by Legrain (1923): “Ninmu son of Balala”; “Kad Nannarzu, scribe, son of Lulamu.” Quantitatively: of the 10,767 distinct transliterated seal legends in the CDLI corpus (see §4.3 and the data appendix), 77% contain *dumu* ‘son of’, 23% the title *dub-sar* ‘scribe’, and 26% *ARAD* ‘servant of’.

⁹From the Wikipedia *List of traded commodities* (accessed 2026-06-11), bundled as `commodities.txt`; letters only, non-alphabetic characters treated as breaks; computed by `compute_rR.py`.

3.2 The behavioral mathematical model

The numerator r is a count of adjacent collisions. Every symbol has a probability $1/V$ of matching its preceding symbol. A length- L sequence has $(L - 1)$ adjacent pairs, each matching with probability $1/V$ under uniform sampling. By linearity of expectation, the number of adjacent repeats for length L is

$$E[r] = (L - 1)/V. \quad (1)$$

The denominator R is a count of any and all collisions of a symbol with any preceding symbol. A symbol at position k counts as a repetition iff it has a collision event with at least one of the $(k - 1)$ symbols at earlier positions. From probability theory, if the probability of an event is P , then the probability of it occurring at least once in n events is $1 - (1 - P)^n$. Therefore when the probability of a collision is $1/V$, the probability that the symbol at position k is a repeat of at least one of the $k - 1$ preceding symbols under uniform sampling is $1 - (1 - 1/V)^{k-1}$. By linearity of expectation, the number of any repeats for length L is

$$E[R] = \sum_{k=1}^L \left[1 - \left(1 - \frac{1}{V}\right)^{k-1} \right] = L - \sum_{j=0}^{L-1} \left(1 - \frac{1}{V}\right)^j \quad (2)$$

This is a finite geometric sum of the form $\sum_{i=0}^n x^i = (1 - x^{n+1})/(1 - x)$ with $n = L - 1$, $x = 1 - 1/V$ and $1 - x = 1/V$, i.e. $V(1 - (1 - 1/V)^L)$, giving

$$E[R] = L - V \left(1 - \left(1 - \frac{1}{V}\right)^L \right). \quad (3)$$

This closed form of R works for all cases mathematically but from an intuitive sense, R behaves differently for corpora of different distribution parameters. While the adjacent repetition count r depends only on its adjacent symbol, the total repetition count R changes based on the values of V and L . When the symbol inventory V is small, the chances of repeats are high even at short string lengths. For $V = 2$, a length of 2 has a collision probability of $1/V = 0.5$. As V gets larger, the length needs to be larger and larger for a collision. For $V = 10000$, randomly drawn symbols may not see any collision for strings of length up to $\sqrt{10000} = 100$ due to the birthday collision paradox. Once the inscription length L grows well past the inventory size V , most of the symbols of V have already appeared, that is to say the string is “saturated” with its symbol inventory and additional symbols are mostly collisions, and therefore not bound by the birthday collision paradox. These two mathematically different behaviors may be termed *unsaturated* and *saturated* regimes.

When $L \gg V$, the string is saturated $(1 - 1/V)^L \rightarrow 0$, and the closed-form reduces to $E[R] \rightarrow L - V$. Essentially, almost every position is a repeat and only V positions are first occurrences.

When $L \ll V$, the string is unsaturated, and $E[R]$ can be approximated by binomial-expanding $(1 - 1/V)^L$ in the closed form above. Since

$$\left(1 - \frac{1}{V}\right)^L = 1 - \frac{L}{V} + \frac{L(L-1)}{2V^2} - \dots,$$

the constant and linear terms cancel against the outer L in $E[R] = L - V(1 - (1 - 1/V)^L)$, leaving $E[R] \approx L(L - 1)/2V$.

The unsaturated approximation may also be derived by pair-counting. A length- L sequence has $\binom{L}{2} = L(L - 1)/2$ unordered position-pairs, each matching with probability $1/V$ under uniform sampling. By linearity of expectation,

$$E[\text{\#matching pairs}] = \frac{L(L - 1)}{2V}.$$

When $L \ll V$, the probability that any specific triple of positions all share a symbol is $1/V^2$, so the expected number of triples $\binom{L}{3}/V^2 \ll \binom{L}{2}/V$. Almost every match is a distinct pair, R equals the number of matching pairs, and

$$E[R] \approx \frac{L(L-1)}{2V} \approx \frac{L^2}{2V} \quad (L \ll V). \quad (4)$$

Combining with $E[r] = (L-1)/V$:

$$E[r/R] \approx \begin{cases} \frac{(L-1)/V}{L(L-1)/(2V)} = \frac{2}{L} & \text{unsaturated regime, } L \ll V \\ \frac{(L-1)/V}{L-V} \approx \frac{1}{V} & \text{saturated regime, } L \gg V. \end{cases} \quad (5)$$

The value $2/L$ in the unsaturated regime inherits the error term from the binomial approximation of its denominator but has no further error added in the division of r/R . Once $E[R]$ is replaced by its unsaturated approximation $L(L-1)/(2V)$, the factors $(L-1)$ and V cancel algebraically and the ratio simplifies to $2/L$ with no further rounding. The disappearance of the inventory size V has an intuitive interpretation: Each repeat is a distinct matching *pair*, equally likely to be any of the $\binom{L}{2}$ position-pairs, of which exactly $L-1$ of those pairs are adjacent. Since both r and R count matching pairs, the common $1/V$ match probability cancels in the ratio, leaving the purely combinatorial fraction of pairs that are adjacent. This value $(L-1)/\binom{L}{2} = 2/L$ is independent of V : r/R measures where the repeats landed, and a short inscription simply has fewer non-adjacent places for them to land.

The behavior of the saturated regime also has an intuitive interpretation: when $L \gg V$ almost every position is a repeat ($R \rightarrow L - V \approx L$) while adjacent positions match with probability $1/V$ ($r \approx L/V$), giving $r/R \rightarrow 1/V$.

This derivation of equation (5) assumes a fixed length L but may be adapted to a corpus of varying length with a mean inscription length of $\bar{L}$. When inscription lengths L_i are unequal, the ratio of the expected pooled r to the pooled R is $2 \sum_i (L_i - 1) / \sum_i L_i (L_i - 1)$,¹⁰ which is at most $2/\bar{L}$: longer inscriptions contribute quadratically to R due to equation (4) but only linearly to r due to equation (1). Length dispersion alone therefore pushes a corpus below the $2/\bar{L}$ curve one would naively draw at its mean length, so a measured r/R below that mean-length curve is not by itself evidence of non-random structure.¹¹

3.3 Distributions with unequal symbol frequency

When the null distribution is modified to have unequal symbol frequencies, the value of the ratio increases above that of its equal distribution counterpart.

For an i.i.d. source with unigram probabilities p_i and coincidence sums $S_k = \sum_i p_i^k$,¹² the unsaturated computation generalizes to $E[r] = (L-1)S_2$ and $E[R] \approx \binom{L}{2}S_2 - \binom{L}{3}S_3$, giving

$$E[r]/E[R] = \frac{2}{L} \left(1 + \frac{(L-2)S_3}{3S_2} + \dots \right) \geq \frac{2}{L},$$

¹⁰RRR is computed as the ratio of summed counts, so its null ratio value is the ratio of summed expectations $\sum_i E[r_i] / \sum_i E[R_i]$. Substituting the unsaturated per-inscription values $E[r_i] = (L_i - 1)/V$ and $E[R_i] \approx \binom{L_i}{2}/V = L_i(L_i - 1)/(2V)$ from Section 3, the V cancels, giving $(\frac{1}{V} \sum_i (L_i - 1)) / (\frac{1}{2V} \sum_i L_i(L_i - 1)) = 2 \sum_i (L_i - 1) / \sum_i L_i(L_i - 1)$, which collapses to $2/L$ when all $L_i = L$.

¹¹Proof of inequality: cross-multiplying gives $\sum_i (L_i - \bar{L})(L_i - 1) \geq 0$, which equals $M \text{Var}(L) \geq 0$ over the M inscriptions.

¹² S_k is the k -fold coincidence sum, which is the probability that k symbols drawn independently from the distribution are all identical. For one *specific* symbol i this is the product p_i^k ; the V choices of common symbol are mutually exclusive, so summing gives $S_k = \sum_i p_i^k$. Hence $S_1 = 1$ and $S_2 = \sum_i p_i^2$ is the two-draw collision probability, and S_3 the three-draw collision probability. For the uniform null distribution $p_i = 1/V$ all V products are equal, giving $S_k = V(1/V)^k = 1/V^{k-1}$. Thus $S_2 = 1/V$, $S_3 = 1/V^2$ and $S_3/S_2 = 1/V$, which turns the general factor $(L-2)S_3/(3S_2)$ into the uniform $(L-2)/(3V)$.

derived in Appendix A, since the bracketed correction is nonnegative; more precisely $S_3/S_2 \geq S_2 \geq 1/V$ by Cauchy–Schwarz,¹³ so the bound $2/L$ is approached only in the uniform large- V limit ($S_3/S_2 = 1/V \rightarrow 0$). Frequency skew therefore can only *raise* the r/R ratio of a null distribution above $2/L$ but never lower it. Any i.i.d. source is above $2/L$, therefore an r/R below $2/L$ cannot come from frequency skew, rather it requires structure that suppresses repetition, while a value above $2/L$ is ambiguous: it could be simply frequency skew or a repetition-enhancing structure or just short length. Therefore it has no evidentiary value.¹⁴

3.4 Threshold interaction

Figure 1 plots the behavior of the r/R ratio under the null distribution as the length changes, against the threshold of 0.10. At short lengths, it is inevitable that the ratio scores closer to the maximally non-linguistic 1.00. As the length increases, the ratio keeps falling until it crosses the threshold into the linguistic zone. In the unsaturated regime, setting the threshold to the asymptote $2/L = 0.10$ gives the point of crossing at $L = 20$ shown by the red circle. Therefore, any text shorter than ~ 20 has an expected $r/R > 0.10$.

The dashed line represents the behavior at $V \rightarrow \infty$. It represents a floor that all i.i.d. distributions share as a limit. At a finite fixed V , the null ratio value exceeds this $2/L$ approximation by a factor¹⁵ of $1 + (L - 2)/(3V) + \dots$, which appears on the chart as a curve that pulls away from the line. Due to the V in the denominator, the lesser the value of V , the further the line moves away. The $V = 20$ solid grey curve moves much further than the $V = 100$ dot-dashed grey curve.

Any finite V thus shifts the null curve towards higher scores, and a falling curve shifted up crosses the threshold line further to the right: the crossing marked by the circle in Figure 1 can only move right of the approximate $L = 20$. Solving the exact closed form for $E[r]/E[R] = 0.10$ places the crossing at $L \approx 21.3$ for $V = 100$ and $L \approx 25.6$ for $V = 30$, approaching 20 only as $V \rightarrow \infty$. Texts shorter than ~ 20 tokens sit above the threshold *a fortiori*. $2/L$ therefore is the lower limit and exceeding the threshold is thus the chance-level default for short texts and requires no decorative structure and is no evidence against linguistic content.

In the saturated regime the threshold becomes $V = 10$. The two asymptotes cross at $L = 2V$ illustrated by Figure 2. The transition between regimes is gradual, and near the crossover neither asymptote is accurate. At $L = 2V$ the exact ratio is $\approx 1.76/V$ for large V , $1.73/V$ at $V = 20$.¹⁶

For distributions with $V \leq 10$, the threshold is never crossed at any length. The null ratio value approaches its saturated floor $1/V \geq 0.10$ from above.

3.5 Empirical validation

The saturated and unsaturated regimes can be demonstrated by simulating any random sequence such as 10,000 rolls of a 20 sided die (D20). The r/R ratio of such inscriptions starts close to the unsaturated regime of $\approx 2/L = 2/10 = 0.2$ and tends to the saturated regime

¹³Draw a random symbol I from the distribution, $\Pr[I = i] = p_i$, where p_i is the probability of the symbol drawn. Its moments are $E[p_I] = \sum_i p_i \cdot p_i = S_2$ and $E[p_I^2] = \sum_i p_i \cdot p_i^2 = S_3$. Hence $S_3 - S_2^2 = \text{Var}(p_I) \geq 0$. Dividing by S_2 gives $S_3/S_2 \geq S_2$, with equality iff p_i is constant, i.e. the uniform distribution. $1 = (\sum_i p_i)^2 \leq V \sum_i p_i^2 = V S_2$ (Cauchy–Schwarz against the all-ones vector), with equality only on the uniform distribution.

¹⁴The correction term $(L - 2)S_3/(3S_2)$ was checked against a simulated Zipfian source with $V = 1000$ and $L = 10$: the simulated ratio sat $\sim 27\%$ above $2/L$, matching the correction term.

¹⁵derived in Appendix A

¹⁶ $L = 2V$ in equations (1) and (3) gives $E[r] = 2 - 1/V$ and $E[R] = V(1 + (1 - 1/V)^{2V})$. Hence $E[r]/E[R] = \frac{2 - 1/V}{V(1 + (1 - 1/V)^{2V})}$. As $V \rightarrow \infty$, $(1 - 1/V)^{2V} \rightarrow e^{-2}$ and the numerator $\rightarrow 2$, the closed-form coefficient $\frac{2}{1 + e^{-2}} = 1.7616\dots$, i.e. $1.76/V$. At $V = 20$ the value is $1.73/V$.

asymptote $1/V = 1/20 \approx 0.05$ with longer rolls of dice as the inscription. The latter value is well below the proposed linguistic threshold of 0.1. In other words, RRR gives a linguistic score even to a random stream if $L \gg V$.

L rolls	L/V	$2/L$ asymptote	$1/V$ asymptote	exact $E[r]/E[R]$	RRR verdict
10	0.5	0.2000	0.050	0.2280	non-linguistic
100	5	0.0200	0.050	0.0618	linguistic
1,000	50	0.0020	0.050	0.0510	linguistic
10,000	500	0.0002	0.050	0.0501	linguistic

Table 1: **RRR of D20 rolls** tending from non-linguistic to linguistic as it approaches its asymptote of $1/V$.

corpus	V	L	L/V	$2/L$	r/R	comment	reason
Sumerian	692	4.4	0.006	0.455	0.67	$> 2/L$	compound signs
Ancient Chinese	701	4.1	0.006	0.488	0.048	$\ll 2/L$	short length
Chinese	2738	13.6	0.005	0.147	0.048	$< 2/L$	
Linear B	220	12.4	0.056	0.161	0.055	$< 2/L$	
Korean (syl.)	984	22.4	0.023	0.089	0.020	$< 2/L$	
Egyptian	691	30.8	0.045	0.065	0.10	$> 2/L$	logograph recurrence
Korean (jamo)	102	43.8	0.43	0.046	0.022	$< 2/L$	
Arabic	62	42.9	0.69	0.047	0.032	$< 2/L$	
Malayalam	80	49.6	0.62	0.040	0.022	$< 2/L$	
English	76	50.3	0.66	0.040	0.035	$\approx 2/L$	
Hindi	77	61.2	0.79	0.033	0.0017	$\ll 2/L$	disjoint unicode
Tamil	61	38.5	0.63	0.052	0.006	$\ll 2/L$	disjoint unicode
Oriya	75	40.2	0.54	0.050	0.0075	$\ll 2/L$	disjoint unicode
Amharic	219	159.9	0.73	0.013	0.018	$\approx 2/L$	
<i>control rows:</i>							
base64 sha128	64	22	0.34	0.091	0.099	$\approx 2/L$	random control
Carlsen PGN (chars)	27	378.7	14.0	0.005	0.0004	$\ll 2/L$	disjoint control

Table 2: **RRR Linguistic comparanda** from Supplementary Table S3, with the unsaturated asymptote $2/L$ attached. Observed values largely sit at or below the null ratio value $2/L$; the three exceptions are Sumerian (compound-sign inflation), Egyptian (logographic recurrence), and Amharic (long L puts it just above). Below the midrule: base64 sha128 is a null distribution control sitting at $2/L$. Carlsen PGN (7,484 of Magnus Carlsen’s tournament games at the PGN character level, not language) sits below every linguistic entry at $r/R = 0.0004$.

The RRR results are presented somewhat like a numerical value of languageness. It is maximally non-linguistic at 1.0 and maximally linguistic at 0.0 with 0.1 as the boundary analogous to pH 7. This is directly stated as such (Sproat, 2014, p. 469):

the five corpora with the highest r/R are non-linguistic (if one counts the Indus system as non-linguistic) and the nine corpora with the lowest values are all linguistic.

Treating RRR as a numerical metric of linguistic nature would mean that Hindi is 20 times more linguistic than English and 400 times more linguistic than Sumerian. These extreme ratios are an artifact of Unicode encoding rather than glyph shape. Devanāgarī consonants 0x0915–0x0939 and vowel signs 0x093E–0x094D live in disjoint Unicode ranges and alternate by phonotactic necessity — adjacent identical codepoints are structurally suppressed. This means that केके when scanned as codepoints, does not register any adjacent repeats, although visually it is a geminated glyph pair. Scanning by visual glyphs instead of codepoints would tighten the range of the reported results.

We measure r/R on random base64 hashes in table 2, and note that the expected r/R ratio for a short random string is $2/L$, which transitions to $1/V$ as the length gets longer. The values reported for most linguistic corpora sit below $2/L$ because they have structure such as phonotactic alternation and disjoint codepoint classes that suppresses repetition relative to the null distribution. Structure can also push values in the opposing direction above the null ratio value, as several non-linguistic comparanda show in table 3.

Non-randomness, whether linguistic or non-linguistic, may show up as a very low score. Magnus Carlsen’s chess games in PGN notation show up as even more linguistic than Hindi, for the same reason: the structure of chess notation rarely allows geminate symbols.

The picture for the non-linguistic corpora is essentially a mirror image. The high scores are a proxy for short inscription length. One of the inferences of this is that given names which often have geminates without distant repeats, like “Matthew” and “Anne”, score maximally non-linguistic. This becomes markedly amplified in Finnish, which uses doubled letters for long vowels. The r/R ratio is highly unstable at short lengths simply because a single gemination classes as maximally non-linguistic, while a single non-geminate repetition classes as maximally linguistic. This is why the reported values range from under 0.1 all the way to 0.86.

corpus	V	L	L/V	$2/L$	r/R	comment	reason
Vinca	185	1.4	0.008	> 1	0.59	$< 2/L$	short L
Barn stars	32	3.1	0.097	0.645	0.86	$> 2/L$	
Pictish stones	104	3.5	0.034	0.571	0.26	$< 2/L$	short L
Weather icons	16	5.0	0.313	0.400	0.79	$> 2/L$	low V
Totem poles	477	5.5	0.012	0.364	0.63	$> 2/L$	
Asian emoticons	334	5.9	0.018	0.339	0.10	$\ll 2/L$	facial symmetry
Kudurrus	64	13.6	0.213	0.147	0.099	$< 2/L$	
<i>Modern linguistic corpora misclassified as non-linguistic by RRR (this paper):</i>							
Top-100 American names	25	6.15	0.246	0.325	0.24	$< 2/L$	short L
Top-100 Finnish names	21	5.72	0.272	0.350	0.53	$> 2/L$	high gemination
<i>Corpus of disputed linguistic status:</i>							
Indus bar seals	209	6.14	0.029	0.326	0.58	$> 2/L$	high gemination
<i>null distribution control at IVC-comparable length:</i>							
6-char hash, base64	64	6.0	0.094	0.333	0.2923	$\approx 2/L$	random control

Table 3: Non-linguistic RRR comparanda tend to the $2/L$ asymptote but are highly unstable. Finnish first names at $r/R = 0.526$ sit in the same band as the Indus bar seals at 0.58. (For Vinča, the null ratio value $2/L$ exceeds 1 and is vacuous since $r/R \leq 1$.)

A compelling case for the instability over short inscription corpora is demonstrated by the station names of the Beijing and Shenzhen subways. Beijing is maximally linguistic while Shenzhen is maximally non-linguistic, and this is determined entirely by non-adjacent repetition in just two Beijing stations while Shenzhen has none. $R = 2$ is due to just two stations having a repeated character: 北苑路北 (“North Beiyuan Road” — 北 at positions 0 and 3) and 北海北 (“North Beihai” — 北 at positions 0 and 2).

Essentially, r/R as a noisy proxy for length was readily acknowledged (Sproat, 2014, p. 471):

“[T]he key insight at work here comes down to a rather simple measure: how long on average are the extant texts in the symbol system? If they are short, then they are probably non-linguistic; if they are long they are probably linguistic.”

The monograph documenting the comparison corpora states it more plainly still: the measures that “do seem useful” — r/R among them — “are useful largely because they are both highly correlated with a rather trivial feature: mean text length” (Sproat, 2013, p. 1).

corpus	# texts	mean L	V	r	R	$\frac{r+\epsilon}{R+\epsilon}$	verdict
Shenzhen metro, Hanzi	308	2.61	273	0	0	1	maximally non-linguistic
Beijing, Hanzi	250	3.22	308	0	2	0	maximally linguistic

Table 4: Shenzhen and Beijing Metro station names. At the script’s native Hanzi unit, Shenzhen’s per-station R is identically zero: each station name uses non-repeating Hanzi glyphs (e.g. 罗湖, 国贸), while Beijing’s $R = 2$ comes from just two stations. Across the 308 Shenzhen stations, the same Hanzi recur prolifically (南 21 times, 园 18, 湾 17) — the actual linguistic signal lives at the cross-inscription level, which per-text scoping discards.

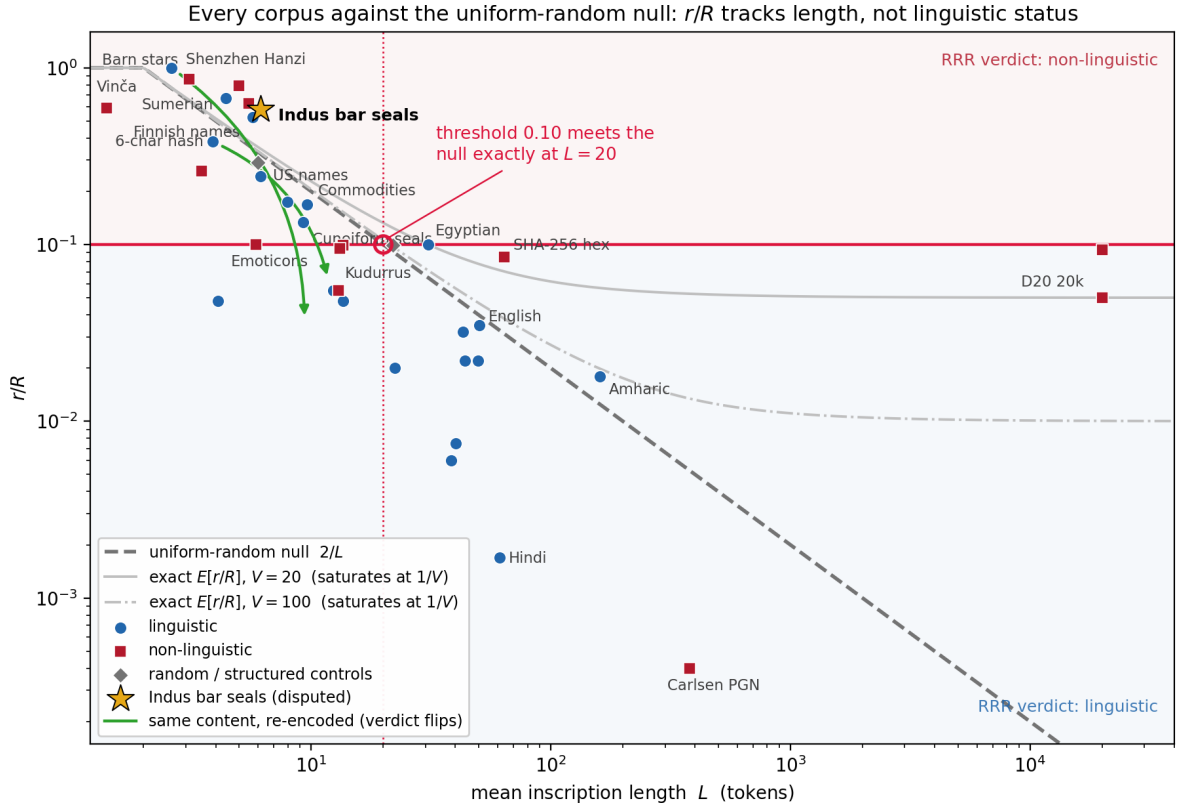


Figure 1: Every corpus in $(L, r/R)$ space against the uniform-random null curve $2/L$ (dashed; the $V \rightarrow \infty$ limit, the floor of the family of finite- V null curves). The RRR threshold $r/R = 0.10$ (horizontal line) meets the null curve exactly at $L = 20$ (circle): the verdict is a length test. At seal-like lengths the classes interleave — Finnish given names sit above totem poles, and the Indus bar seals (star) sit between the two. Green arrows re-encode identical content (Shenzhen Hanzi $\rightarrow$ pinyin; CDLI Sumerian glyphs $\rightarrow$ letters) and flip the verdict. The grey curves are the exact $E[r/R]$ for $V = 20$ (solid) and $V = 100$ (dash-dotted): each sits above the $2/L$ limit — the closer, the larger its V — before saturating at its own floor $1/V$. The D20 dice stream rides the $V = 20$ curve through the threshold into the “linguistic” region in the saturated regime. Hindi and Carlsen PGN (bottom) both sit far below the null curve — structured suppression of repetition, linguistic and non-linguistic alike.

Sproat (2014, p. 469) in fact anticipated the length confound, reporting a Pearson correlation of -0.49 between r/R and mean text length and arguing from a one-way ANOVA ($F = 5.75$, $p = 0.027$ for mean length vs. $F = 15.83$, $p = 0.0008$ for r/R) that the results “cannot be simply reduced to length differences.” But a linear correlation is the wrong functional form for a hyperbolic relationship.¹⁷ Recomputing on the same values (the 21 Table 2 corpora plus the Indus bar seals; the rounded published figures give Pearson -0.48 against his reported -0.49), the Pearson correlation of r/R against the null curve $\min(2/L, 1)$ rises to 0.73 , and the rank (Spearman) correlation between r/R and mean length — which is free of any functional-form assumption — is -0.82 . The modest -0.49 is an artifact of fitting a straight line to a $2/L$ curve; the length dependence is far stronger than the linear test suggested.

Figure 1 draws the picture in full: every corpus in this paper plotted against the $2/L$ null curve, with the threshold geometry, the class interleaving at seal-like lengths, the encoding-flip arrows, and the saturated-regime dice stream in a single frame. We have mathematically reduced RRR to $2/L$, explained the reasons for the instability of its values, and verified that the reported values are in the expected theoretical range compared to a random control and falsification control. We next explain in detail how these instabilities present themselves.

4 Failure modes

4.1 Encoding fidelity

In the corpora underlying RRR, the alphabetic and syllabic linguistic corpora were tokenized with spaces “represented as a separate token (#)” (Sproat, 2013, §7), which feeds the non-adjacent repetition count R . No delimiter tokens were used for the non-linguistic corpora — including barn stars, which are nearly universally spaced apart by the width of at least one symbol.¹⁸ The monograph’s own exhibit makes the asymmetry concrete: slide AR(2)F of the Farrell collection (apparently the ca.-1940 Berks County barn photographer William Ferrell, whose images are reproduced in Donmoyer, 2019) shows a barn face bearing three hex signs separated by several symbol-widths of blank wall — entire doors intervene — yet its encoding is three adjacent tokens on a single line (WHEEL_OF_FORTUNE 20_POINT_STAR WHEEL_OF_FORTUNE), with no delimiter (Sproat, 2013, p. 100). An English sentence whose words were separated by proportionally equal gaps would have received a # token at every boundary. If a space is counted as in the linguistic corpora, this corpus, which was classed highly non-linguistic, becomes maximally linguistic. This is an important consideration for bronze age scripts, which were in use long before word spacing became conventional: classical antiquity wrote in *scriptura continua*, and systematic word separation in the Latin West is a medieval scribal innovation (Saenger, 1997). A metric that conflates these two groups loses reliability.

4.2 Encoding-depth

The same underlying content gives different r/R values under different tokenization-unit choices. This falsifies any claim of determining linguistic status for an unknown script whose choice of tokenization unit is itself a hypothesis.

¹⁷Pearson’s coefficient measures how well points fit a *straight line* and not how strongly one variable depends on another. When synthetic data is placed exactly on the null curve $r/R = \min(2/L, 1)$ at the actual mean lengths of these 22 corpora with zero noise, so that the dependence on length is total and deterministic, the linear test returns Pearson -0.58 against L , while Pearson against the null curve returns 1.0 and Spearman returns -0.999 (two corpora tie at $L = 13.6$). A coefficient near -0.5 is thus what a pure $2/L$ relationship produces at these lengths. It is no evidence that the length dependence is weak.

¹⁸See the façade photographs and descriptions in Donmoyer (2019): stars are placed singly across the forebay siding and gable ends, crowning individual doors and windows. For example, “four twelve-pointed stars across the barn’s forebay siding” (p. 3), with architectural features intervening between the symbols.

corpus	V	L	r	R	r/R
barn-star-like (raw symbols, no delim.)	5	5.07	442	499	0.886
same, one space inserted between symbols	6	9.13	0	1112	0.000

Table 5: A synthetic barn-star corpus matched to RRR’s reported $r/R = 0.86$ for barn stars. Adding a single space token between every adjacent symbol (encoding the physical gap that exists on the wall) drops r/R from 0.886 to 0.000: an 89-percentage-point flip from “maximally non-linguistic” to “maximally linguistic.” RRR applies this delimiter inclusion to English but not to barn stars.

We can demonstrate this using different encodings of the same 128 random bits (e.g. MD5, or SHA-256 truncated to its first 128 bits) encoded in four standard display bases. These versions are classed differently, indicating high instability due to variation in encoding depth. Table 6 demonstrates how with increasing encoding depth, the RRR classes the same data differently. Base64 is in the unsaturated regime and is essentially equal to the $2/L = 0.091$ asymptote. The binary encoding is deeply saturated ($L/V = 64$), with observed $r/R = 0.503$ essentially equal to the $1/V = 0.500$ asymptote. All four sit close to their null distributions (exact $E[r]/E[R]$: 0.101, 0.107, 0.131, 0.504); the digit streams are not perfectly i.i.d. uniform — leading-digit and padding effects in the decimal and base64 forms — but the deviations are small.

encoding	V	L	L/V	$2/L$	$1/V$	observed r/R
base64	64	22	0.34	0.091	0.016	0.099
hex	16	32	2.00	0.063	0.063	0.108
decimal	10	38.6	3.86	0.052	0.100	0.126
binary	2	128	64	0.016	0.500	0.503

Table 6: Identical 128 random bits encoded in four number-bases. r/R traverses the range [0.099, 0.503] on fixed random content.

The base64 form classifies as linguistic by RRR (just below the 0.10 threshold), while the hex, decimal, and binary forms all classify as non-linguistic. The encoding choice alone produces both verdicts on the same 128 random bits, and the four rows make the unsaturated-to-saturated regime sweep explicit on a single content type.

Whereas the random control is a reference demonstration of the impact of the chosen encoding on the RRR verdict, attested cases can vary dramatically based on length and actual depth of encoding. In many cases, the score swings from one maximal point to the other, purely on encoding depth.

corpus / encoding	# texts	mean L	V	$\frac{r+\epsilon}{R+\epsilon}$	verdict
Sumerian, glyph level	30,120	3.88	762	0.382	non-linguistic
Sumerian, letter level	30,120	11.87	31	0.061	linguistic
Shenzhen Metro, Hanzi	308	2.61	273	1.000	non-linguistic
Shenzhen Metro, pinyin	308	9.41	26	0.036	linguistic
Shuffled cards ($A\spadesuit, 2\clubsuit$)	100 decks	52.00	52	1.000	non-linguistic
Shuffled cards ($A, \spadesuit, 2, \clubsuit$)	100 decks	104.00	17	0.000	linguistic

Table 7: Identical underlying content is classed differently based on encoding depth. Encoding defines V and L , which changes the result.

These observations have a theoretical basis. When V is very small, such as in binary strings or Morse code, the count of r adjacent repetitions increases dramatically, while high V decreases

R total repetitions dramatically. The sweet spot between them is determined by the tension between the two asymptotes: $1/V$ in the saturated regime (small V , large L) and, writing N for the information content in bits so that a base- V encoding has length $L = N/\log_2 V$, $2/L = 2\log_2 V/N$ in the unsaturated regime (large V , small L). As V increases the saturated asymptote falls but the unsaturated asymptote rises. The two cross at the regime boundary $V \cdot \log_2 V = N/2$, and the exact $E[r/R]$ achieves its minimum just past that crossover.

Figure 2 traces the tension between the two asymptotes as a plot of the ratio $E[r/R]$ vs inventory size V for fixed bits of information N . For $N = 128$ bits, the minimum value of the ratio sits at $V \approx 32$ (the RFC 4648 base32 alphabet) with $E[r/R] \approx 0.099$ — *just below RRR’s 0.10 threshold*. As content size grows, the minimum drops. Cryptographic randomness, at the right encoding base, is arbitrarily strongly classified as linguistic by RRR.

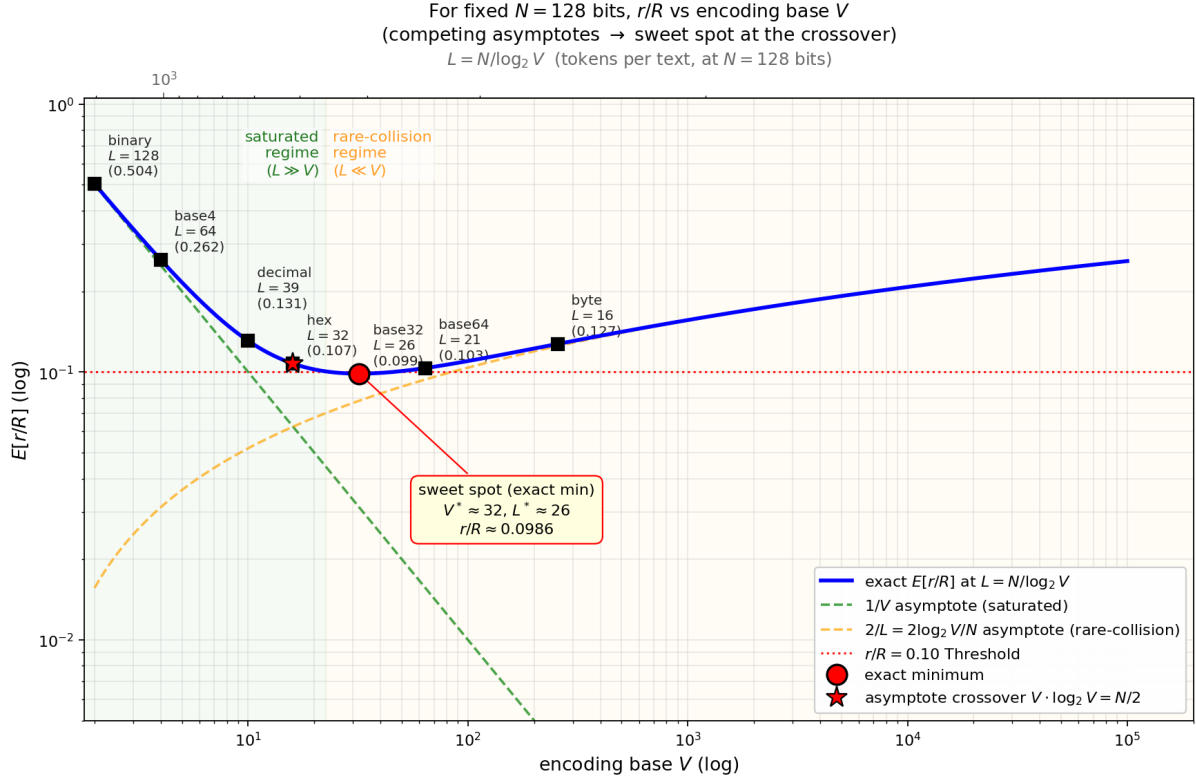
4.3 The threshold collapse

The r/R measure is reported as a point estimate with no confidence interval and no error bar. Sumerian, a bronze age writing system for example, scores more non-linguistic at 0.67 than the Indus bar seals, potentially another bronze age script, at 0.58. There is no way to be certain about any value, and only with a priori knowledge can we justify a compliant value or excuse a non-compliant one.

The missing error bars can be supplied for the corpora whose raw data accompany this paper. Resampling inscriptions with replacement within each corpus (nonparametric bootstrap, $B \geq 500$ replicates, 95% percentile intervals) gives Table 8. At seal-corpus sizes the intervals are wide — the American-names interval spans ± 0.10 , and the commodity-label interval straddles the threshold itself — while large corpora are tight. Any argument that turns on whether a 100-inscription corpus scores 0.53 or 0.60 is resting on sampling noise: at these corpus sizes, r/R cannot bear a binary verdict even on its own terms. The ICIT rows add the tokenization sensitivity of §4.2, measured on the Indus corpus itself: with doubled-sign codes counted as single tokens, with those codes expanded into their constituent glyphs, and in Sproat’s Mahadevan-based bar-seal corpus, the same script scores 0.26, 0.70, and 0.58 respectively — a spread covering most of the “non-linguistic” band, produced entirely by edition and tokenization choices.

corpus	n	r/R	95% CI
Top-100 American names	100	0.243	[0.152, 0.347]
Top-100 Finnish names	100	0.526	[0.448, 0.612]
Commodity labels	58	0.169	[0.092, 0.268]
CDLI Mesopotamian seal legends	10,749	0.134	[0.127, 0.142]
SHA-256(1..100) hex	100	0.085	[0.077, 0.092]
SHA-256(1..100) base64	100	0.050	[0.038, 0.063]
English dictionary, by word	43,220	0.174	[0.171, 0.177]
ICIT Indus all, as coded	4,640	0.264	[0.195, 0.333]
ICIT Indus all, doubled signs expanded	4,640	0.704	[0.659, 0.750]
ICIT Indus seals, doubled signs expanded	2,200	0.630	[0.565, 0.695]

Table 8: Nonparametric bootstrap 95% percentile intervals for r/R , resampling inscriptions with replacement within each corpus (computed by `compute_rR.py`; the ICIT rows require the ICIT sign-sequence export plus the bundled mapping `icit_xlits.csv` of the 31 doubled/quadrupled sign codes, and treat damage and space markers as sequence breaks; $B = 2000$ replicates, $B = 500$ for the dictionary, fixed seed, ϵ -regularized ratio). For census-like lists (top-100 names, the full dictionary) the interval measures compositional sensitivity at the given corpus size rather than sampling error from a larger population. Single-sequence corpora (D20, 3-D6 streams) have no inscription-level resampling unit and are omitted.



Sweet-spot V^* sits near the regime boundary $V \cdot \log_2 V = N/2$.
 Larger N pushes the optimum to higher V and lower r/R .

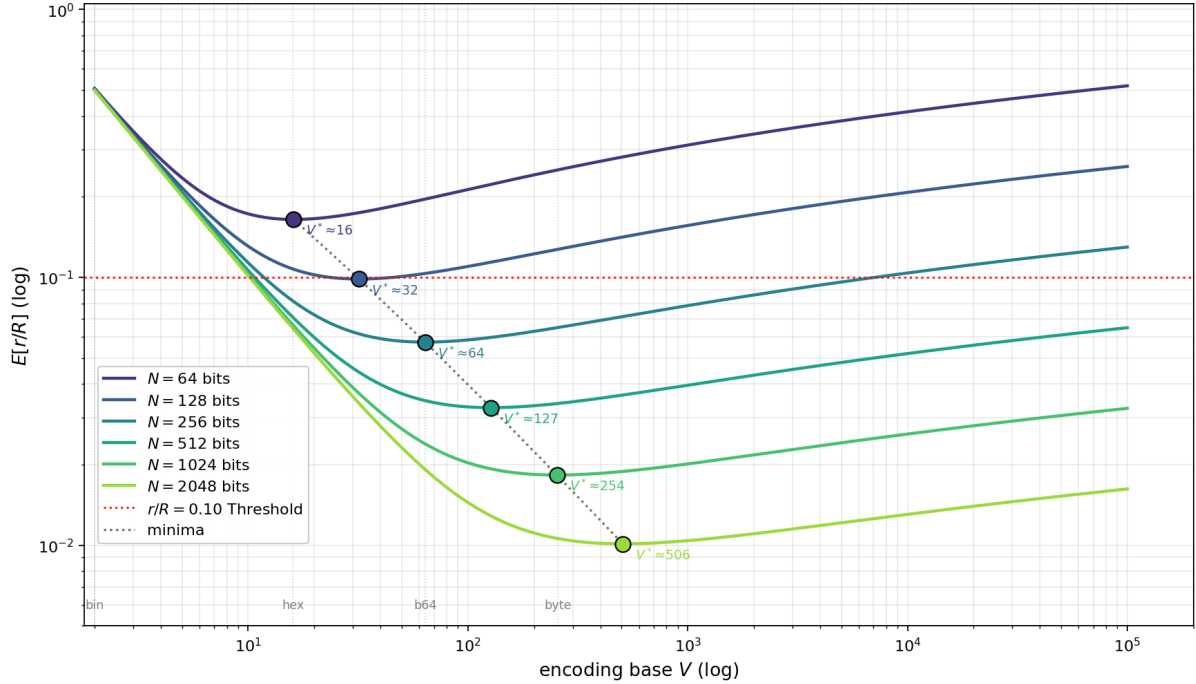


Figure 2: The r/R encoding-base sweet spot. $N = \text{bits of information content}$ (e.g. $N = 128$ for a 128-bit hash). For a chosen encoding base V , the resulting text length is $L = N/\log_2 V$ tokens. **Top panel:** Asymptote crossover for fixed content $N = 128$ bits, $E[r/R]$ as a function of encoding base V (blue). The asymptote crossover (red star, at $V \cdot \log_2 V = N/2 \approx V = 16$) is close to but slightly left of the exact minimum (red circle, at $V \approx 32, E[r/R] \approx 0.099$). **Bottom panel:** family of $E[r/R]$ curves for $N \in \{64, 128, 256, 512, 1024, 2048\}$ bits. The sweet-spot trajectory (dotted) connects the exact minima.

In the length-truncation experiment of Sproat (2014, p. 469), every text was trimmed to its first six symbols and each corpus to its first 500 texts; the separation was reported as “not as clean” as before, and the CART trees classifying the Indus corpus as non-linguistic fell from 0.81 to 0.63 mean held-out accuracy when the truncated ratio replaced the original (Sproat, 2014, p. 472). This indicates an extreme sensitivity to inscription length, making the ratio unsuitable for any short-text-dominated corpus.

The RRR threshold of $\tau = 0.10$ works very well for the selected corpora. However, as we have seen the linguistic and non-linguistic corpora are essentially bifurcated by length and therefore a metric that is a proxy for length has no predictive power therein.

When we break the bifurcation by adding corpora with short linguistic and long non-linguistic characteristics, the resulting r/R values are not bimodal between linguistic and non-linguistic classes anymore. Linguistic content spans $r/R \in [0.0017, 1.000]$ and non-linguistic content spans $r/R \in [0.0000, 0.86]$, with the modes overlapping across the threshold. The threshold value 0.10 no longer separates the classes.

We can quantify RRR’s discriminative power directly as a binary classifier using Youden’s J statistic, the standard sensitivity-plus-specificity summary measure of a thresholded test:

$$J = \text{sensitivity} + \text{specificity} - 1, \quad J \in [-1, +1]$$

where $J = 1$ corresponds to perfect classification, $J = 0$ to a classifier indistinguishable from chance (the ROC diagonal), and $J = -1$ to a perfectly-wrong classifier (one whose predictions would become perfect if flipped). Computing J at RRR’s chosen threshold $\tau = 0.10$, we find that J collapses from 0.857 to -0.03 : on a corpus set that includes its natural failure modes, RRR’s sensitivity-plus-specificity at its own threshold is no better than chance. The argument that falsified Rao’s bigram entropy claim, applied to RRR, falsifies RRR.

scenario	total	TP	FN	FP	TN	sens	spec	$J(\tau = 0.10)$
Sproat’s unambiguous corpora	18	12	2	0	4	0.86	1.00	0.857
+ 13 counter-examples	31	12	9	6	4	0.57	0.40	−0.029

Table 9: Youden’s J for RRR at its chosen threshold $\tau = 0.10$. The unambiguous set comprises the 14 linguistic and 4 of the 7 non-linguistic comparanda of Sproat (2014); Vinča, Pictish stones, and the kudurrus are excluded as contested. The two false negatives are Sumerian (0.67) and Egyptian (0.10). Adding the thirteen counter-examples of Table 10 collapses sensitivity to 0.57 and specificity to 0.40, giving $J = -0.03$, at or below chance.

The 13 counter-example corpora that collapsed J from 0.857 to below zero are itemized in Table 10: seven linguistic corpora that RRR classed as non-linguistic (false negatives), and six non-linguistic corpora that RRR classed as linguistic (false positives). The Shenzhen row depends on the ϵ convention for its $r = R = 0$ edge case; excluding it as undefined instead of scoring it leaves $J = 0.000$ exactly, so the collapse does not hinge on that convention.

Specificity falls from 1.00 to 0.40 (a 0.60 drop), while sensitivity falls from 0.86 to 0.57 (a 0.29 drop). RRR’s curated non-linguistic comparanda gave it a perfect specificity of 1.00. When we added ISBNs, IP addresses, cryptographic hashes, random dice rolls and other structured non-linguistic content, specificity fell below 0.50. Sensitivity loss causes RRR to fail on linguistic content at short L on corpora that are characteristic of ancient seal inscriptions: Finnish names, Sumerian glyphs, Hanzi station names, commodity labels, cuneiform seal legends.

While $\tau = 0.10$ clearly fails as a universal threshold, we can investigate whether *any* threshold is viable by testing the area under the ROC curve (AUC). This is the probability that a randomly chosen linguistic corpus could score below a randomly chosen non-linguistic one, evaluated over all possible thresholds (Mann-Whitney). Table 11 compares r/R against the trivial classifier “mean inscription length L ” on three corpus sets. On the calibration set, raw

corpus	actual class	mean L	V	r/R	RRR verdict (error)
<i>False negatives: linguistic content classified non-linguistic</i>					
Top-100 American given names	linguistic	6.15	25	0.243	non-linguistic (FN)
Top-100 Finnish given names	linguistic	5.72	21	0.526	non-linguistic (FN)
Exchange-traded commodity names	linguistic	9.62	24	0.169	non-linguistic (FN)
English dictionary, by word	linguistic	7.95	29	0.174	non-linguistic (FN)
Sumerian glyph level (CDLI Gudea)	linguistic	3.88	762	0.382	non-linguistic (FN)
Shenzhen Metro, Hanzi per-station	linguistic	2.61	273	1.000	non-linguistic (FN)
CDLI Mesopotamian seal legends	linguistic	9.28	1032	0.134	non-linguistic (FN)
<i>False positives: non-linguistic content classified linguistic</i>					
Top Shelf Comix ISBN-13 list	non-linguistic	13.0	10	0.055	linguistic (FP)
2ip.io top-100 IPv4, dotted /24	non-linguistic	13.2	11	0.095	linguistic (FP)
SHA-256(1..100) hex	non-linguistic	64.0	16	0.085	linguistic (FP)
3-D6 sum (sic bo), long sequence	non-linguistic	$\gg 1$	16	0.093	linguistic (FP)
D20 uniform-random rolls	non-linguistic	$\gg 1$	20	0.050	linguistic (FP)
Carlsen PGN, character level	non-linguistic	378.7	27	0.0004	linguistic (FP)

Table 10: The thirteen counter-example corpora (twelve modern, one ancient) that collapse Youden’s J from 0.857 (RRR’s unambiguous calibration set) to -0.03 (at or below chance) when added to the corpus. The Shenzhen Hanzi row uses the ϵ -regularized convention $(r+\epsilon)/(R+\epsilon) = 1$ for the $r = R = 0$ edge case. Net effect on the 31-corpus confusion matrix: TP/FN/FP/TN = 12/9/6/4, sensitivity 0.57, specificity 0.40, $J = -0.03$.

length alone achieves AUC 0.893, within 0.06 of r/R ’s 0.955: nearly all of RRR’s advertised discriminative power was already available from L , with no repetition statistics at all. When all 31 corpora with known linguistic status are analyzed in their native encoding, i.e., hashes in hexadecimal and dictionary words in the English alphabet, r/R falls to 0.619. (one-sided permutation $p = 0.15$, 20,000 label shuffles). Adding or removing corpora in different encodings moves the AUC between 0.60 and 0.69 and the p -value from 0.20 to 0.047. The statistical significance level moving between conventionally significant and random chance simply by altering the corpora and encoding choices alone is itself a demonstration of requirement 5 (§5): the sampling frame. Without a sampling frame, such p -values are an artifact of the selected corpora and not the metric’s validity.

On the eleven corpora at the lengths typical of ancient seal and label inscriptions, $L \in [2.5, 8]$ the AUC falls to 0.643 with $p = 0.27$, showing no detectable advantage over chance. Within this band no threshold cleanly separates the classes: Asian emoticons score below every name list, while Sumerian and the Shenzhen station names score above the totem poles.

corpus set	ling/non	AUC(r/R)	perm. p	AUC(L)
Sproat calibration set (unambiguous)	14/4	0.955	0.002	0.893
Full set (31 corpora)	21/10	0.619	0.15	0.443
Length-restricted band $L \in [2.5, 8]$	7/4	0.643	0.27	0.500

Table 11: Threshold-free discrimination. AUC(r/R): probability that a random linguistic corpus has lower r/R than a random non-linguistic one; AUC(L): the same for mean inscription length (higher = linguistic); perm. p : one-sided permutation p -value for AUC(r/R) > 0.5 under 20,000 label shuffles. On the calibration set, length alone recovers nearly all of r/R ’s performance. Within the length-restricted band (Sumerian, Ancient Chinese, barn stars, weather icons, totem poles, Asian emoticons, American and Finnish given names, per-word English, CDLI Sumerian glyphs, Shenzhen Hanzi), r/R is indistinguishable from chance.

The thirteen corpora additions are not exotic adversarial constructions. Given names, dic-

tionary words, station names, commodity labels, deciphered seal legends, ISBNs, IP addresses, hashes, chess game records, and dice rolls are large, natural, high-prevalence classes of symbol sequences that are arguably more representative of real-world short inscriptions and identifier systems than kudurrus or totem poles.

Dice pips are a pictographic numeral convention and dice are used decoratively in mosaics exactly as barn stars and weather icons are used ornamentally. Moreover, Sproat’s own project was ambitiously open-domain. A growing database of non-linguistic corpora including mathematical equations and European heraldry was described as under development (Sproat, 2014, p. 459). These formal notation systems are similar to our corpora such as ISBN and IPv4.

A threshold whose validity depends on excluding the most common modern classes of short linguistic text and long non-linguistic identifiers fails as a discriminant of linguistic status.

A related objection concerns the choice of segmentation: the modern linguistic counterexamples score names and station signs as separate texts, while the dice and hash streams run long. We fix the scoping rule independently of outcome as simply one text per production unit. The symbol sequence occupying one physical carrier such as a station signboard or a printed ISBN or one uninterrupted generation episode such as a hash output. This is the same rule that makes Sproat’s comparanda short: one kudurru or one totem pole for example, and his linguistic corpora long: one running-text line. Consistency requires accepting or rejecting per-name and per-kudurru scoping together. The segmentation choice is not an artifact of our corpora but a structural degree of freedom of RRR itself (§4.2). A stream of dice rolls has no objective production unit and must be picked arbitrarily. This leads to the inevitable effect of the length choice. Table 1 illustrates how the same generating process flips the result on different stream lengths. Accordingly, the J and AUC figures above should be read as demonstrations that large natural classes of counterexamples exist, not as estimates of RRR’s performance on a defined population of symbol systems.

The RRR metric is unfalsifiable as a language test in the Popperian sense. The counterexamples presented such as the Finnish names of ISBN identifiers can be explained *post hoc* by the corpus having short text length or by having a small/large inventory or by encoding choice. These happen to be exactly the properties the structural analysis identifies. The metric measures an inscription’s length and sign inventory rather than its linguistic nature.

This metric seems to have been created without a random control and without a length control. Hence, the threshold worked for only the length bifurcated corpora. The chosen threshold did not come from any theoretical basis and was reported without confidence intervals. The failure of the threshold when exposed to a larger variety of corpora was inevitable.

5 Requirements for a valid discriminant

The failure modes documented above are not specific to r/R . They are the generic ways a corpus statistic comes to masquerade as a linguistic test. An ideal discriminant function of linguistic nature would need to satisfy the following requirements.

1. **Null hypothesis/Random control.** A string of random symbols should score maximally non-linguistic regardless of length. All other systems would have some informational content, which should result in a lower non-linguistic score. Linguistic corpora should score close to maximally linguistic.
2. **Length invariance.** String subsets *within* the same corpus should score reasonably close to each other regardless of length. The error term or confidence level may change to reflect the measurement uncertainty of short strings.
3. **Encoding invariance.** Changing the encoding should not vary the scores greatly. Ideally, the expected variation is pre-defined. For a completely undeciphered system, this requires

a mathematical justification of the proposed symbol inventory, which is a non-trivial task to accomplish objectively.

4. **Threshold clarity.** Linguistic and non-linguistic systems should cluster far from each other with a gap that is wider than the cluster. That way, the threshold has a comfortable buffer of certainty. If the error bars overlap the threshold (Table 8), the metric is not credible.
5. **Sampling frame.** Qualify symbol systems and the corpus applicability criteria before scores are computed. In addition, the scoring should be validated against corpora not used to create the metric. The results need to be stable and predictable over asymmetrically large number of corpora that were not used to create the metric.
6. **Boundary definition and stability.** Define behavior at boundary conditions and degenerate values ($R = 0$). Also define under per-inscription vs. pooled scoping before evaluation. Each of these conventions moved at least one corpus across RRR’s threshold. (§4.3).

RRR fails on all six requirements, which are indeed demanding. For deciphered systems with externally validated thresholds the first, third, and fourth can reasonably be relaxed. An undeciphered system has unknown tokenization depth and fidelity and its nature remains an open question until partial or full decipherment. Such a system’s nature cannot be reasonably determined unless all requirements are satisfied.

6 Conclusion

The r/R measure rests on three implicit premises, each tied to one of its three components. We have shown that all three fail empirically. We have shown that the ratio is a proxy for length ≥ 20 . We have empirically shown that the ratio fails for long non-linguistic texts such as chess PGN and short linguistic texts such as Finnish and American given names and 10,749 deciphered Mesopotamian seal legends. We have shown that random strings are classed purely on length, even with zero linguistic content. We have shown that its calibration-set performance does not survive corpora that break the length bifurcation: at its chosen threshold its sensitivity-plus-specificity falls to chance level (Youden’s $J = -0.03$), and within the length-restricted band the threshold-free AUC shows no detectable advantage over chance. We have shown how encoding depth can flip the results on identical texts.

None of this establishes that the Indus symbols are writing. Invalidating RRR removes one argument that they are not; the question itself remains open.

RRR is not a discriminant of linguistic vs. non-linguistic data, for the same reason that bigram conditional entropy and block entropy are not: all three are measures of structural regularity, and many non-linguistic processes (compression, fixed prefixes, encoding-depth alternation, run-length coding, random sources at moderate L/V) produce structural regularity indistinguishable from language.

It may be impossible to create a mathematical model that distinguishes linguistic vs non-linguistic systems without a priori knowledge because there is no proven universal model for known and unknown languages, let alone a universal model of orthographic systems.

A Finite- V correction to the $2/L$ null ratio value

The unsaturated null ratio value $E[r]/E[R] \rightarrow 2/L$ derived in Section 3 is exact in the limit $V \rightarrow \infty$ at fixed L . At finite V , the exact $E[r]/E[R]$ exceeds $2/L$ by the factor $1 + (L - 2)/(3V) + \dots$. This appendix derives that factor and its i.i.d. generalization. Note that this

correction factor only applies to the unsaturated regime, since the saturated regime does not use this approximation.

Start from the exact expectations of Section 3:

$$E[r] = \frac{L-1}{V}, \quad E[R] = L - V \left(1 - \left(1 - \frac{1}{V} \right)^L \right). \quad (6)$$

Binomial-expand $(1 - 1/V)^L$ in powers of $1/V$,

$$\left(1 - \frac{1}{V} \right)^L = 1 - \frac{L}{V} + \frac{\binom{L}{2}}{V^2} - \frac{\binom{L}{3}}{V^3} + \dots, \quad (7)$$

$$E[R] = L - V \left(1 - \left(1 - \frac{1}{V} \right)^L \right) = L - V \left(1 - \left(1 - \frac{L}{V} + \frac{\binom{L}{2}}{V^2} - \frac{\binom{L}{3}}{V^3} + \dots \right) \right), \quad (8)$$

$$E[R] = L - V + V \left(1 - \frac{L}{V} + \frac{\binom{L}{2}}{V^2} - \frac{\binom{L}{3}}{V^3} + \dots \right), \quad (9)$$

$$E[R] = L - V + V - V \cdot \frac{L}{V} + V \cdot \frac{\binom{L}{2}}{V^2} - V \cdot \frac{\binom{L}{3}}{V^3} + \dots, \quad (10)$$

$$E[R] = \frac{\binom{L}{2}}{V} - \frac{\binom{L}{3}}{V^2} + \dots = \frac{L(L-1)}{2V} \left[1 - \frac{L-2}{3V} + \dots \right], \quad (11)$$

using $\binom{L}{3}/\binom{L}{2} = (L-2)/3$. An intuitive way to understand this is that $\binom{L}{2}/V$ counts pairs and $\binom{L}{3}/V^2$ subtracts the overcounted triples which share a symbol with the already counted pairs. This is extended to matches beyond triples using the inclusion-exclusion rule.¹⁹ The bracketed factor therefore, is the correction factor to $E[R] \approx L(L-1)/2V$. The $(L-1)$ and V cancel when we divide $E[r] = (L-1)/V$ by $E[R] \approx L(L-1)/2V$.

$$\frac{E[r]}{E[R]} = \frac{2}{L} \cdot \frac{1}{1 - \frac{L-2}{3V} + \dots} = \frac{2}{L} \left(1 + \frac{L-2}{3V} + \dots \right), \quad (12)$$

the last step by $1/(1-\epsilon) = 1 + \epsilon + \dots$ with $\epsilon = (L-2)/(3V)$.

When the frequencies are unequal, $1/V$ needs to be replaced by the coincidence sums. For a general i.i.d. source with unigram probabilities p_i and coincidence sums $S_k = \sum_i p_i^k$, a specific pair of positions matches with probability $\sum_i p_i^2 = S_2$. The pair-match probability is a sum over the V symbols because a match does not fix *which* symbol is shared among the two positions. The events “both are symbol i ” have probability p_i^2 and are mutually exclusive across the V symbols, so by the law of total probability $\Pr[\text{match}] = \sum_i p_i^2 = S_2$. The string length plays no role here, since it enters only through the *number* of pairs: $L-1$ adjacent pairs for r and $\binom{L}{2}$ total position-pairs for R . A specific j -tuple all-matches with probability $\sum_i p_i^j = S_j$, so by linearity of expectation

$$E[r] = (L-1)S_2, \quad E[R] = \sum_{j \geq 2} (-1)^j \binom{L}{j} S_j = \binom{L}{2} S_2 - \binom{L}{3} S_3 + \dots, \quad (13)$$

¹⁹The alternating series comes from inclusion-exclusion. Fix position b and let $A_a = \{s_a = s_b\}$ for each earlier position $a \in \{1, \dots, b-1\}$; then b is a repeat iff $\bigcup_a A_a$ occurs. A k -fold intersection $\bigcap_{a \in I} A_a$ ($|I| = k$) forces the k positions of I *plus* b , i.e., $k+1$ positions, to share one symbol, an event of probability S_{k+1} for every such k -subset by i.i.d. exchangeability. Inclusion-exclusion then gives $\Pr[b \text{ a repeat}] = \sum_{k=1}^{b-1} (-1)^{k-1} \binom{b-1}{k} S_{k+1}$, and summing over $b = 1, \dots, L$ with the hockey-stick identity $\sum_{m=0}^{L-1} \binom{m}{k} = \binom{L}{k+1}$ yields $E[R] = \sum_{j \geq 2} (-1)^j \binom{L}{j} S_j$ (set $j = k+1$).

the alternating series being the general counterpart of the binomial expansion above, which reduces to it via $S_j = 1/V^{j-1}$, i.e. $\binom{L}{j}/V^{j-1}$. Divide, and factor $\binom{L}{2}S_2$ out of the denominator:

$$\frac{E[r]}{E[R]} = \frac{(L-1)S_2}{\binom{L}{2}S_2 \left[1 - \frac{\binom{L}{3}}{\binom{L}{2}} \frac{S_3}{S_2} + \dots \right]}. \quad (14)$$

In the prefactor S_2 cancels and $\frac{L-1}{\binom{L}{2}} = \frac{2}{L}$; in the bracket $\frac{\binom{L}{3}}{\binom{L}{2}} = \frac{L-2}{3}$. Hence

$$\frac{E[r]}{E[R]} = \frac{2}{L} \cdot \frac{1}{1 - \frac{L-2}{3} \frac{S_3}{S_2} + \dots} = \frac{2}{L} \left(1 + \frac{(L-2)S_3}{3S_2} + \dots \right), \quad (15)$$

the last step by $1/(1-\epsilon) = 1 + \epsilon + \dots$ with $\epsilon = (L-2)S_3/(3S_2)$. This reduces to the uniform result via $S_2 = 1/V$, $S_3 = 1/V^2$, $S_3/S_2 = 1/V$. The omitted terms $(\dots)$ are higher order in the unsaturated small parameter ($\sim L/V$ in the uniform case). They collect the truncated $\binom{L}{4}S_4 - \dots$ tail of $E[R]$ together with the ϵ^2 and higher terms of $1/(1-\epsilon)$, the leading one being $\frac{(L-2)^2}{9}(S_3/S_2)^2 - \frac{(L-2)(L-3)}{12}(S_4/S_2)$ — and are negligible for $L \ll V$.

It is clear from the fact that the correction term is positive for every $L > 2$ (all $S_k > 0$) that the exact i.i.d. null ratio value never falls below $2/L$. A finite V and frequency skew both *raise* it. The uniform null distribution's value is the floor by Cauchy–Schwarz $S_2^2 \leq S_3$ and $1/V \leq S_2$, hence $S_3/S_2 \geq 1/V$. Also, since the exact $E[r]/E[R]$ sits *above* $2/L$, the exact form's threshold-equivalent length is provably larger than the approximate form's $L^* = 20$. The conclusion that sub-20-symbol texts clear the $r/R = 0.10$ threshold therefore holds *a fortiori*, and under any i.i.d. unigram distribution rather than only the uniform one. The magnitude is verified using a Zipfian source at $V = 1000$, $L = 10$ sitting $\sim 27\%$ above $2/L$ in simulation, matching $(L-2)S_3/(3S_2)$.

Data and code availability

The companion archive is permanently deposited at <https://doi.org/10.5281/zenodo.21175888> (all versions: <https://doi.org/10.5281/zenodo.20651448>). It contains the paper source, the figure-generation scripts (`rR_sweetspot.py`, `rR_scatter.py`), the rendered figures (`rR_sweetspot.png`, `rR_scatter.png`), and a consolidated reproducibility script (`compute_rR.py`) that recomputes every numeric r/R value, Youden's J value, AUC and permutation p -value, bootstrap interval, gemination rate, and sweet-spot V^* reported in the paper. Run `python3 compute_rR.py` to verify. The script depends only on the Python standard library (`hashlib`, `random`, `base64`, `urllib`, `zipfile`). The companion archive bundles `american.txt` (a 43,220-word SCOWL American-English dictionary, used for the English-dict row of Table 10 and the Morse re-encoding reported by `compute_rR.py`) and the name lists `ssa_top100.txt` and `dvv_top100.txt`; the Carlsen control row downloads `Carlsen.zip` from <https://www.pgnmentor.com/players/Carlsen.zip> on first run. PGN Mentor updates that collection as new games are played, so the script pins the copy used here (accessed 2026-06-11: 7,484 games, SHA-256 `de327354...4e2062`); when the live file has drifted from the pin, the script instead fetches the archived pinned copy at <https://archive.org/download/Carlsen/Carlsen.zip>. All datasets referenced in this paper are listed below with their sources. Where the dataset is computed (random sampling, synthetic generation), the algorithm is in `compute_rR.py` and the section's stated parameters.

Linguistic and non-linguistic comparanda (Sproat 2014, Table S3)

Numerical r/R , mean text length, and inventory-size values for the 21 corpora in Sproat's Supplementary Table S3 are reproduced from Sproat (2014) and his online Supplementary

Materials at <https://muse.jhu.edu/journals/language/v090/90.2.sproat01.pdf>. These include: Sumerian, Egyptian, Chinese, Ancient Chinese (oracle bones), English, Arabic, Korean (syllable), Korean (jamo), Linear B, Hindi, Malayalam, Tamil, Oriya, Amharic, Vinča symbols, Barn stars, Pictish stones (argued to be linguistic by Lee, Jonathan, and Ziman, 2010a), Weather icons, Totem poles, Asian emoticons, and Mesopotamian deity symbols (kudurrus).

Linguistic corpora used as modern counter-examples

- **Top-100 American given names:** the top-50 male and top-50 female names from the US Social Security Administration “Top names of the last 100 years” table (1926–2025 window, accessed 2026-06-11), bundled as `ssa_top100.txt`; <https://www.ssa.gov/oact/babynames/decades/century.html>.
- **Top-100 Finnish given names:** the top-50 men’s and top-50 women’s names (all name positions) from the first-name statistics file of Finland’s Digital and Population Data Services Agency (DVV), `etunimitilasto-2026-02-03-dvv.xlsx` via <https://www.avoindata.fi/data/fi/dataset/none>, bundled as `dvv_top100.txt`.
- **ICIT Indus corpus** (disputed status): sign sequences from the Interactive Corpus of Indus Texts (Wells & Fuls), <https://www.indus.epigraphica.de>; used for the Indus gemination rate and the bootstrap intervals in Table 8.
- **Sumerian (CDLI Gudea):** reproduced from the Cuneiform Digital Library Initiative ATF dump, <https://github.com/cdli-gh/data>, filtered on `primary_publication` containing `rime 3` and lines containing `gu3-de2-a`. Yields 30,120 lines.
- **Shenzhen Metro station names:** 308 stations from English Wikipedia, *List of Shenzhen Metro stations*, https://en.wikipedia.org/wiki/List_of_Shenzhen_Metro_stations.
- **Beijing Subway station names:** 250 unique stations from English Wikipedia, *List of Beijing Subway stations*, https://en.wikipedia.org/wiki/List_of_Beijing_Subway_stations.
- **English dictionary:** American-English word list, 43,220 entries, from the public-domain SCOWL distribution; used at per-word scoping for the encoding-flip row in Table 10.
- **Commodity labels:** 58 traded-commodity contract names from the English Wikipedia *List of traded commodities* (accessed 2026-06-11), bundled as `commodities.txt`.
- **CDLI Mesopotamian seal legends:** all `@seal` blocks extracted from the CDLI ATF dump (`cdliatf_unblocked.atf`, <https://github.com/cdli-gh/data>, accessed 2026-06-11), deduplicated on normalized text: 10,767 distinct legends (10,749 with at least one legible sign), bundled as `cdli_seal_legends.txt` with CDLI attribution.

Non-linguistic corpora and synthetic sources

- **Top Shelf Comix ISBN-13 list:** 207 ISBN-13 codes from <https://www.topshelfcomix.com/catalog/isbn-list>.
- **2ip.io top-100 IPv4 addresses:** 25,600 addresses (top-100 routing prefixes /24-expanded) from <https://2ip.io/analytics/popular-ips/>.
- **Cryptographic hashes:** SHA-256, SHA-256 truncated to 128 bits, and SHA-512 computed via Python `hashlib` over integer inputs $1, \dots, N$, encoded in base64, hex, decimal, and binary as described in §4.2 and Table 6.
- **Shuffled playing cards:** 100 independent random shuffles of a 52-card deck, generated via Python `random.shuffle`; used in the encoding-flip rows of Table 7.
- **Uniform random dice (D20):** simulated in Python; results reported in Table 1.

- **Magnus Carlsen tournament games (PGN at character level):** 7,484 games from PGN Mentor, <https://www.pgnmentor.com/players/Carlsen.zip> (auto-downloaded by `compute_rR.py` on first run). See Table 2.

Fonts used in this paper (all embedded)

- Latin/math body: Latin Modern + Computer Modern (TeX Live bundled).
- Devanāgarī glyphs: Kohinoor Devanagari (Apple, macOS default).
- Hangul glyphs: Apple SD Gothic Neo (Apple, macOS default).
- Hanzi (Simplified Chinese) glyphs: PingFang SC (Apple, macOS default).
- Japanese (Hiragana, Katakana, Kanji) glyphs: Hiragino Sans (Apple, macOS default).
- Cuneiform glyphs: Noto Sans Cuneiform (Google Noto family, shipped with macOS Supplemental fonts).

References

- Ashtadhyayi.com. 2024. *Aṣṭādhyāyī: Digital Edition of Pāṇini’s Grammar with Sūtra Text and Numbering*. Online edition, accessed 2026-06-11. <https://ashtadhyayi.com>.
- Donmoyer, Patrick J. 2019. *Hex Signs: Sacred and Celestial Symbolism in Pennsylvania Dutch Barn Stars*. Exhibition catalogue, Glencairn Museum, Bryn Athyn, PA, and Pennsylvania German Cultural Heritage Center, Kutztown University.
- ePSD2 Project. 2017. *The Electronic Pennsylvania Sumerian Dictionary, 2nd edition*. Online database, University of Pennsylvania, hosted on ORACC; accessed 2026-06-11. <https://oracc.museum.upenn.edu/epsd2/>.
- Farmer, Steve, Richard Sproat, and Michael Witzel. 2004. “The Collapse of the Indus-Script Thesis: The Myth of a Literate Harappan Civilization.” *Electronic Journal of Vedic Studies* 11 (2): 19–57.
- Inkelas, Sharon. 2014. “Non-Concatenative Derivation: Reduplication.” In *The Oxford Handbook of Derivational Morphology*, edited by Rochelle Lieber and Pavol Štekauer. Oxford: Oxford University Press.
- Lee, Rob, Philip Jonathan, and Pauline Ziman. 2010a. “Pictish Symbols Revealed as a Written Language through Application of Shannon Entropy.” *Proceedings of the Royal Society A* 466 (2121): 2545–2560.
- Lee, Rob, Philip Jonathan, and Pauline Ziman. 2010b. “A Response to Richard Sproat on Random Systems, Writing, and Entropy.” *Computational Linguistics* 36 (4): 791–794.
- Legrain, Leon. 1923. “Some Seals of the Babylonian Collections.” *The Museum Journal* 14 (2). <https://www.penn.museum/sites/journal/1129/>.
- Monier-Williams, Monier. 1899. *A Sanskrit-English Dictionary*. Oxford: Clarendon Press. Digitization (c=mw) served at <https://www.shrauta.com>.
- Rao, Rajesh P. N., Nisha Yadav, Mayank N. Vahia, Hrishikesh Joglekar, R. Adhikari, and Iravatham Mahadevan. 2009. “Entropic Evidence for Linguistic Structure in the Indus Script.” *Science* 324 (5931): 1165.
- Rao, Rajesh P. N. 2010. “Probabilistic Analysis of an Ancient Undeciphered Script.” *IEEE Computer* 43 (4): 76–80.

- Rao, Rajesh P. N., Nisha Yadav, Mayank N. Vahia, Hrishikesh Joglekar, Ronojoy Adhikari, and Iravatham Mahadevan. 2010. “Entropy, the Indus Script, and Language: A Reply to R. Sproat.” *Computational Linguistics* 36 (4): 795–805.
- Saenger, Paul. 1997. *Space Between Words: The Origins of Silent Reading*. Stanford: Stanford University Press.
- Sproat, Richard. 2010a. “Ancient Symbols, Computational Linguistics, and the Reviewing Practices of the General Science Journals.” *Computational Linguistics* 36 (3): 585–594.
- Sproat, Richard. 2010b. “Reply to Rao et al. and Lee et al.” *Computational Linguistics* 36 (4): 807–816.
- Sproat, Richard. 2013. *Corpora and Statistical Analysis of Non-Linguistic Symbol Systems*. Monograph, 21 January 2013. <https://richardsproat.com/monograph.pdf>.
- Sproat, Richard. 2014. “A Statistical Comparison of Written Language and Nonlinguistic Symbol Systems.” *Language* 90 (2): 457–481. <https://doi.org/10.1353/lan.2014.0031>.
- The Unicode Consortium. 2024. *The Unicode Standard, Version 16.0.0*. South San Francisco, CA: The Unicode Consortium. <https://www.unicode.org/versions/Unicode16.0.0/>.